

INDIAN INSTITUTE OF TECHNOLOGY INDORE

DEPARTMENT OF MECHANICAL ENGINEERING

ME 224: Fluid Mechanics

**Numerical Study of Concentration
Gradient Generation in
Y-Junction Microfluidic Devices**

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Abstract

In this project, two inlet streams carrying the same base fluid (water) but different solute (O_2) concentrations are introduced into a Y-junction device. Using computational fluid dynamics (CFD), the study analyzes how diffusion creates a concentration gradient across the channel width and height. In microfluidic systems, mixing occurs mainly through molecular diffusion, as turbulent mixing mechanisms are entirely absent due to the low Reynolds numbers inherent to micro-scale flows.

1 Introduction

In microfluidic systems, mixing occurs mainly through molecular diffusion since turbulence is absent. Y-junction microchannels are commonly used to generate concentration gradients, where two inlet streams merge and interact. In microscale channels, flow is always laminar due to the characteristically low Reynolds number, making diffusion the primary driver for species transport.

2 Literature Review

Y-junction microchannels are widely used for diffusion-based gradient generation. CFD simulations are commonly utilized to model laminar flow and species transport accurately. Gradient formation heavily depends on the Peclet number and inlet flow conditions. While channel geometry is known to influence mixing behavior and gradient development, there are limited studies comprehensively comparing junction angles under varying concentration gradients.

3 Motivation and Objectives

3.1 Motivation

There is a pressing need for controlled gradient generation due to its critical importance in biomedical and chemical applications, such as drug delivery and cell sorting. Furthermore, thorough numerical optimization is required before physical fabrication to minimize costs and maximize efficiency.

3.2 Objectives

- To numerically simulate concentration gradient formation in a Y-junction microchannel.
- To analyze the effect of inlet concentration differences at varying diffusion coefficients, channel heights, and widths.
- To study the influence of the junction merging angle (60° vs 90°).
- To evaluate gradient profiles using localized mass fraction plots.

4 Methodology

4.1 Geometric Design and Dimensions

The 2D Y-junction microchannel geometry was created using CAD software and imported into ANSYS for CFD simulation.

- **Inlet channels:** Width = $25\ \mu\text{m}$, Length = $100\ \mu\text{m}$.
- **Main mixing channel:** Width = $25\ \mu\text{m}$, Length = $150\ \mu\text{m}$.
- **Angles between inlets:** Two cases evaluated (90° and 60°).

4.2 Governing Equations

The numerical simulation is based on the principles of mass and momentum conservation, coupled with species transport. The governing equations for steady, incompressible, and laminar flow are as follows:

1. **Continuity Equation:** Ensures the conservation of mass within the fluid domain.

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

where $\mathbf{u}$ is the velocity vector field.

2. **Navier-Stokes (Momentum) Equation:** Governs the velocity field. Because of the micro-scale dimensions, viscous forces heavily dominate over inertial forces.

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u} \quad (2)$$

where ρ is the fluid density, p is the pressure, and μ is the dynamic viscosity.

3. **Convection-Diffusion (Species Transport) Equation:** Describes the transport of the solute concentration (C). Since there is no turbulence, mixing occurs solely via molecular diffusion and advection.

$$\mathbf{u} \cdot \nabla C = D \nabla^2 C \quad (3)$$

where C is the mass fraction (or concentration) of the species and D is the molecular diffusion coefficient.

4.3 Boundary Conditions and Parametric Cases

We simulated the interface where the two concentrations meet to measure the uniformity and stability of the resulting gradient. The defining parameters include:

- Diffusion coefficient (d_1) = $1 \times 10^{-8}\ \text{m}^2/\text{s}$.
- Diffusion coefficient (d_2) = $1 \times 10^{-11}\ \text{m}^2/\text{s}$.
- Inlet velocities = $0.001\ \text{m/s}$ (both inlets).
- Dynamic viscosity = $0.001\ \text{kg}/(\text{m}\cdot\text{s})$.
- Concentration cases: $C_1 = (0, 1)$ and $C_2 = (0.7, 0.3)$.
- Junction angles: $A_1 = 90^\circ$ and $A_2 = 60^\circ$.

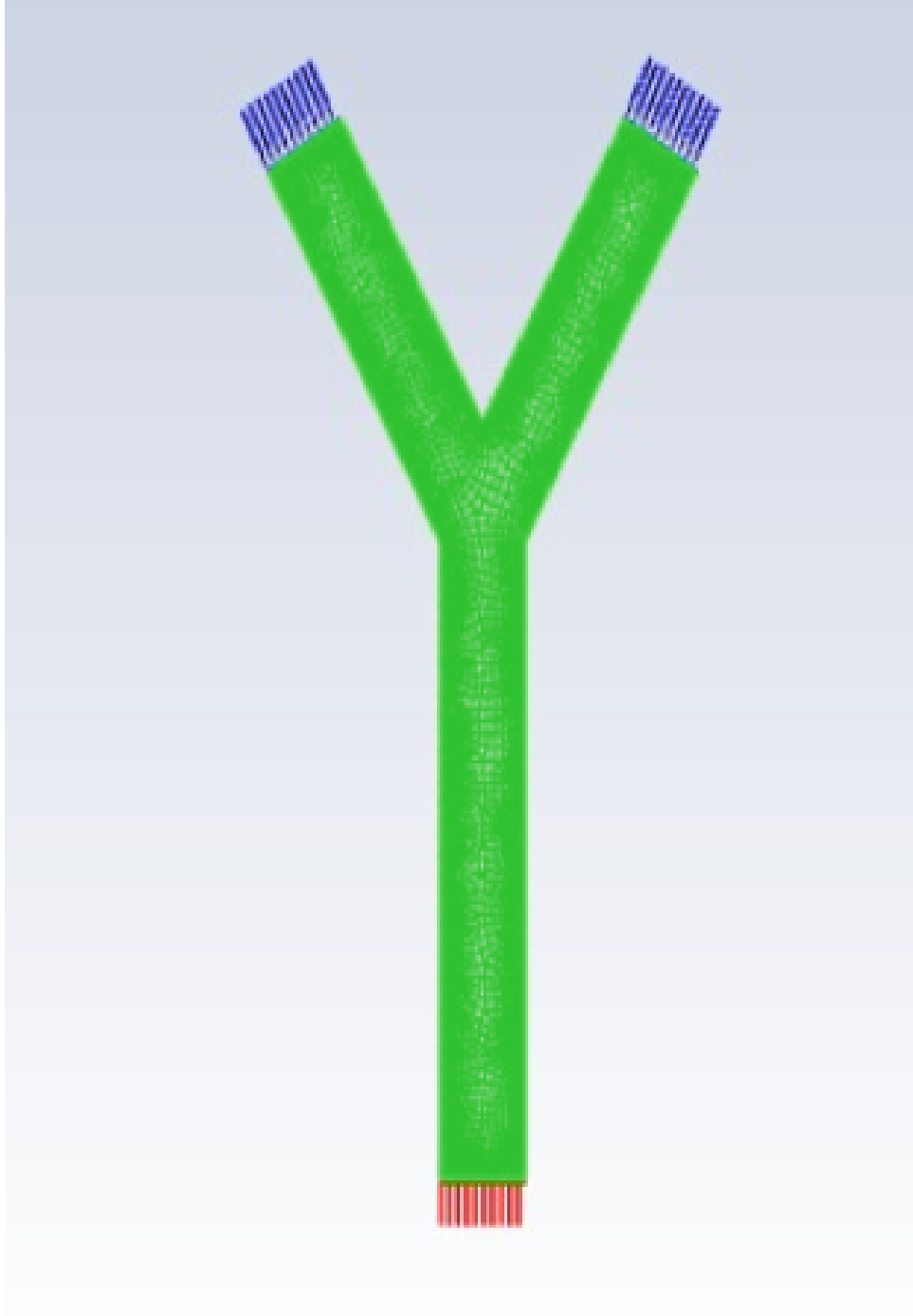


Figure 1: Schematic of the Y-Junction Microchannel Geometry.

5 Results and Discussions

5.1 Gradient Formation at Different Concentrations

Case C_1 (0, 1): Gradient formation is sharper and steeper when the inlet concentration difference is higher. The 90° junction produces a more symmetric and stable interface between the two streams due to equal merging direction, leading to smoother gradient development. The 60° junction shows a slightly shifted and thinner interface, indicating enhanced convection effects and a directional bias in gradient spreading.

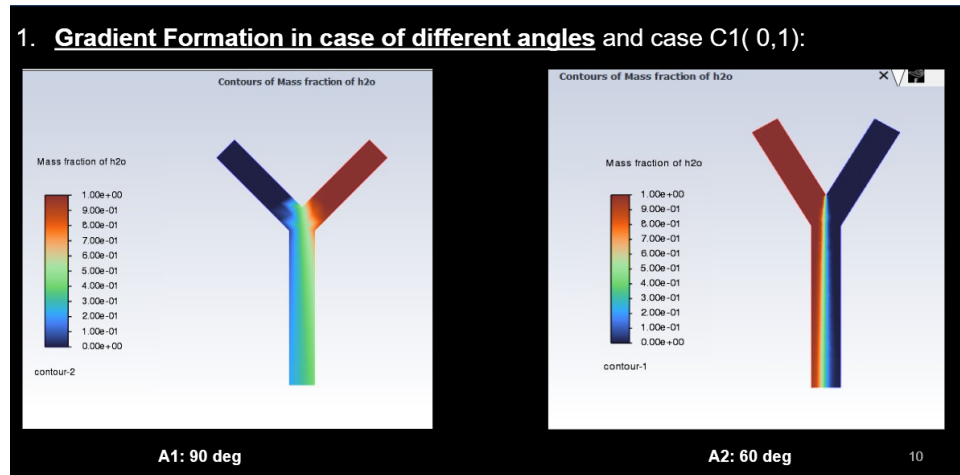


Figure 2: Mass fraction contours for Case C_1 (0, 1) and C_2 (0.7, 0.3).

Case C_2 (0.7, 0.3): For a smaller concentration difference, the gradient becomes smoother and less steep, as the driving diffusion force is proportionately lower.

5.2 Comparison Between Diffusion Coefficients

A higher diffusion coefficient (d_2) produces a wider and smoother concentration gradient, indicating faster molecular diffusion between the streams. A lower diffusion coefficient (d_1) results in a sharper and thinner interface, meaning convection dominates and mixing is weaker. The influence of the diffusion coefficient is most visible near the junction region.

5.3 Gradient Formation at Different Heights

- **At the junction:** The concentration interface is sharp and narrow, as the two streams have just merged and diffusion time is minimal.
- **$50\ \mu\text{m}$ below the junction:** The gradient widens and smooths out due to increasing diffusion time along the channel length.
- **Near the outlet:** The concentration profile is highly uniform, indicating progressive downstream mixing.

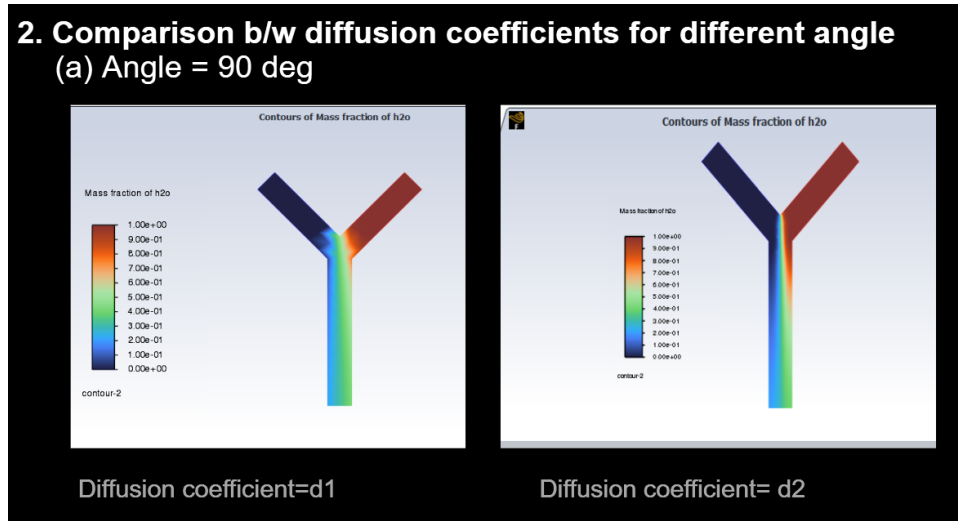


Figure 3: Effect of diffusion coefficients on gradient thickness and mixing.

5.4 Velocity and Pressure Distribution

The velocity profile confirms laminar flow behavior, indicated by smooth, symmetric contours. Velocity peaks in the center of the channel and approaches zero near the walls (no-slip condition), forming a parabolic profile. The pressure contour reveals a gradual, stable decrease from inlet to outlet, confirming steady pressure-driven flow essential for controlled gradient generation.

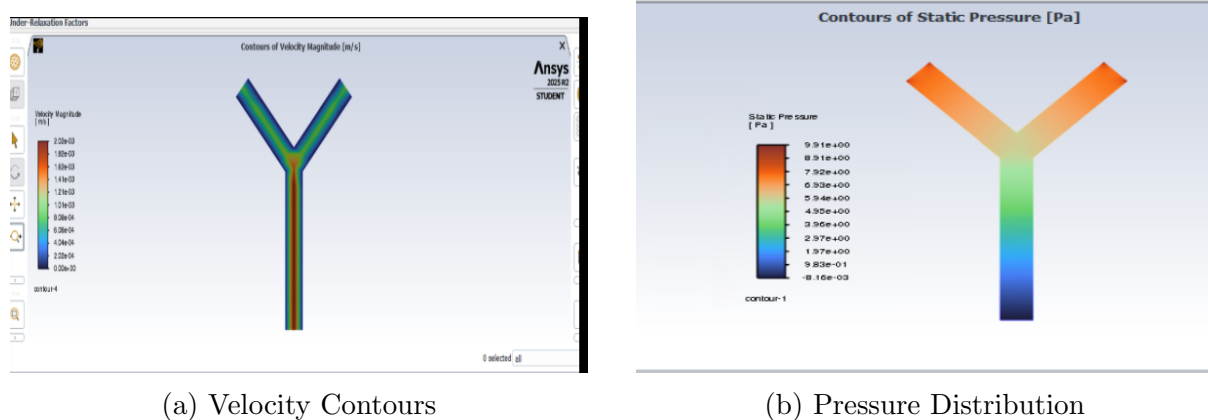


Figure 4: Velocity and Pressure profiles within the microchannel.

6 Conclusion

In this numerical study, the effects of various parameters were analyzed under laminar flow conditions:

- **Concentration:** A larger difference (C_1) yields a sharp, strong gradient, while a smaller difference (C_2) yields a smoother, weaker gradient.
- **Diffusion & Peclet Number:** Faster diffusion (lower Peclet number) promotes

uniform mixing. Lower diffusion (higher Peclet number) preserves a distinct interface longer.

- **Junction Angle:** The 60° configuration facilitates smoother stream interaction, whereas the 90° angle produces a more abrupt interface.
- **Downstream Evolution:** Continuous molecular diffusion ensures the concentration profile becomes progressively smoother and more stable as it travels downstream.

7 Future Scope

- Study additional junction angles and geometric variations to identify the optimal configuration for maximum gradient stability.
- Extend the current 2D simulation to a full 3D model to capture wall-effects more accurately.
- Perform experimental validation by fabricating a physical Y-junction microfluidic device and comparing empirical data with these numerical results.

8 References

1. Diouf, M. M. L. *Journal of Theoretical and Applied Mechanics* 55(1):317.
2. Department of Mechanical Engineering, National Taiwan Science and Technology University. DOI: 10.3390/mi13020338.